

Sourena Khanzadeh

Cognitive Trust Architect

AI Research | Agent Reliability | Causal Evaluation | Privacy

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PROFESSIONAL PROFILE

Computer Science Ph.D. and AI research engineer building auditable agents through causal evaluation, grounded retrieval, tool orchestration, and privacy controls. Research spans **7 peer-reviewed papers and 6 preprints**, including AAAI and Canadian AI. Experience across industry, government, and academia.

PROFESSIONAL EXPERIENCE

AI Engineer, Agentic Systems

May 2026 – Present

Flybits | Full-time | Toronto

- Build agents that plan and execute tasks across internal and partner APIs, with decision traces that support end-to-end audits.
- Ground responses in structured customer-context graphs, combining retrieval and learned models with symbolic constraints.
- Evaluate task completion, groundedness, faithfulness, and recovery from tool failures; use the results to tune model routing for capability, latency, and cost.

AI Research Engineer, sAlpien

May 2026 – Present

MIT Media Lab | Part-time research affiliation

- Prototype perspective-aware agents and private long-term memory using adaptive knowledge graphs (Chronicles); contribute to HCI² benchmarks for transparency and human oversight.

Postdoctoral Fellow / AI Research Engineer

May 2026 – Present

Toronto Metropolitan University | Part-time academic appointment

- Build Python and PyTorch evaluation systems with execution traces, counterfactual replay, and automated scoring to assess agent reasoning, orchestration, and reliability.

Machine Learning Engineer (Graduate Research)

Jan 2023 – Apr 2026

Toronto Metropolitan University | Toronto

- Developed systems for multi-agent coordination, heuristic planning, reinforcement learning, and synthetic-data generation; co-authored research at **AAAI 2026** and **Canadian AI 2024**.
- Built training and inference pipelines with PyTorch, Hugging Face, scikit-learn, and XGBoost; used Docker and GitHub Actions to make experiments repeatable.

ML Engineer / Research Scientist Intern

Jan 2024 – Jan 2025

National Research Council Canada | Toronto

- Developed knowledge-informed anomaly-detection prototypes for imbalanced telemetry, using domain features to address limited labelled data.
- First-authored a journal study of domain knowledge in deep learning for threat defense and co-authored research on auxiliary knowledge for cyberattack detection.

EDUCATION

Ph.D., Computer Science

Sep 2023 – May 2026

Toronto Metropolitan University | **GPA: A+**

Bachelor of Computer Science

Sep 2018 – Dec 2022

Toronto Metropolitan University | **Dean's List, 2020–2021**

A+ in Machine Learning, Artificial Intelligence, Reinforcement Learning, and Computer Vision.

RESEARCH & ENGINEERING ACHIEVEMENTS

Project Ariadne | Causal auditing of LLM agents

Sole author | Preprint, 2026

Designed and built an intervention-and-replay harness to test whether an agent's stated reasoning affects its answer. Identified faithfulness violations in **23 of 30 audited trajectories (76.7%)** under the study's counterfactual protocol.

GasZero | Fuzz-tested smart-contract optimization

First author | IEEE COMPSAC, 2026

Combined static analysis, LLM rewrites, and differential fuzzing. The evaluation across **59 contracts and 527 functions** reported **56.6 million gas** in cumulative savings and 3.99% average execution savings.

Heuristic search | Planning and automated refactoring

AAAI, 2026 | Preprint, 2023

Co-developed a restarting-random-walk planner that was faster or uniquely successful on **70.3% of agile-track tasks with unbounded heuristic regions** versus standard enforced hill-climbing. In [Opti Code Pro](#), reduced node expansions by **82%** and runtime by **74%** versus uninformed A* at full refactoring aggression.

GANsemble | Learning from small, imbalanced datasets

Co-author | Canadian AI, 2024

Co-developed augmentation search and a conditional-GAN pipeline for a 210-image microplastics dataset. Augmentation selection achieved **91.5% mean classifier accuracy** over 10 runs: +4 percentage points versus duplication and +5.5 versus no oversampling.

Interventional Separation Selection | Causal certification

First author | Preprint, 2026

Co-authored a method for certifying agreement across candidate causal models under bounded interventions. On a colored-MNIST causal-abstraction task, required **13.6 interventions per image on average**, with guarantees conditional on the true abstraction belonging to the candidate set.

Provenance Preserving Chronicles | Private AI context

First author | Preprint, 2026

Co-authored a federated disclosure protocol that limits shared context to authorized evidence, preserves provenance, and requires holder approval before releasing raw artifacts; specified the architecture and threat model.

AgentMesh & Folio | Agent orchestration and evaluation

Independent projects | 2025–2026

Created AgentMesh, a Python prototype coordinating planning, coding, debugging, and review with execution-driven repair. Built [Folio](#), an experimental paper-review application with PDF ingestion, live rescoring, and a reproducible classifier-evaluation workflow.

PLLM+ | Reproducible Python dependency repair

Co-author | Preprint, 2026

Co-authored a replay-and-repair pipeline that resolved **1,500/2,891** benchmark snippets versus 1,169 for the baseline, reducing average runtime from **368.7 to 71.8 seconds**. Historical configuration replay supplied 1,495 fixes; LLM fallback supplied five.

TECHNICAL EXPERTISE

AI & evaluation LLM agents, RAG, tool calling, multi-agent orchestration, context and memory, counterfactual evaluation, guardrails, failure analysis, heuristic search, reinforcement learning.

Languages & ML Python, C/C++, TypeScript, Java; PyTorch, Hugging Face, scikit-learn, XGBoost; fine-tuning, GANs, synthetic data.

Engineering Docker, Kubernetes, AWS, PostgreSQL, Redis, Git, GitHub Actions, APIs, CI/CD, reproducible experimentation.

ENTREPRENEURSHIP & EARLIER EXPERIENCE

Founder, Ariadne Growth Systems. Builds measurement, automation, and customer-acquisition systems for service businesses.

Software Engineer Intern, NTN Company

Aug – Sep 2022

Delivered an internal administration portal using Angular, Node.js, Firebase, and Sass.